

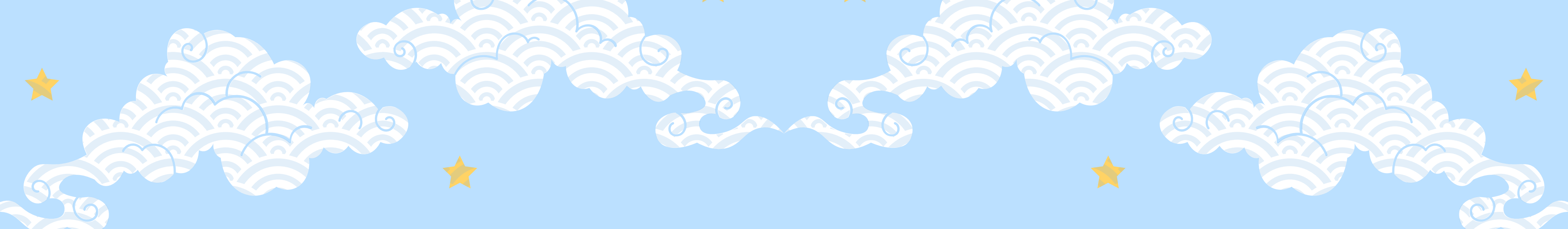
SMART HOUSE SYSTEM

Using Arduino



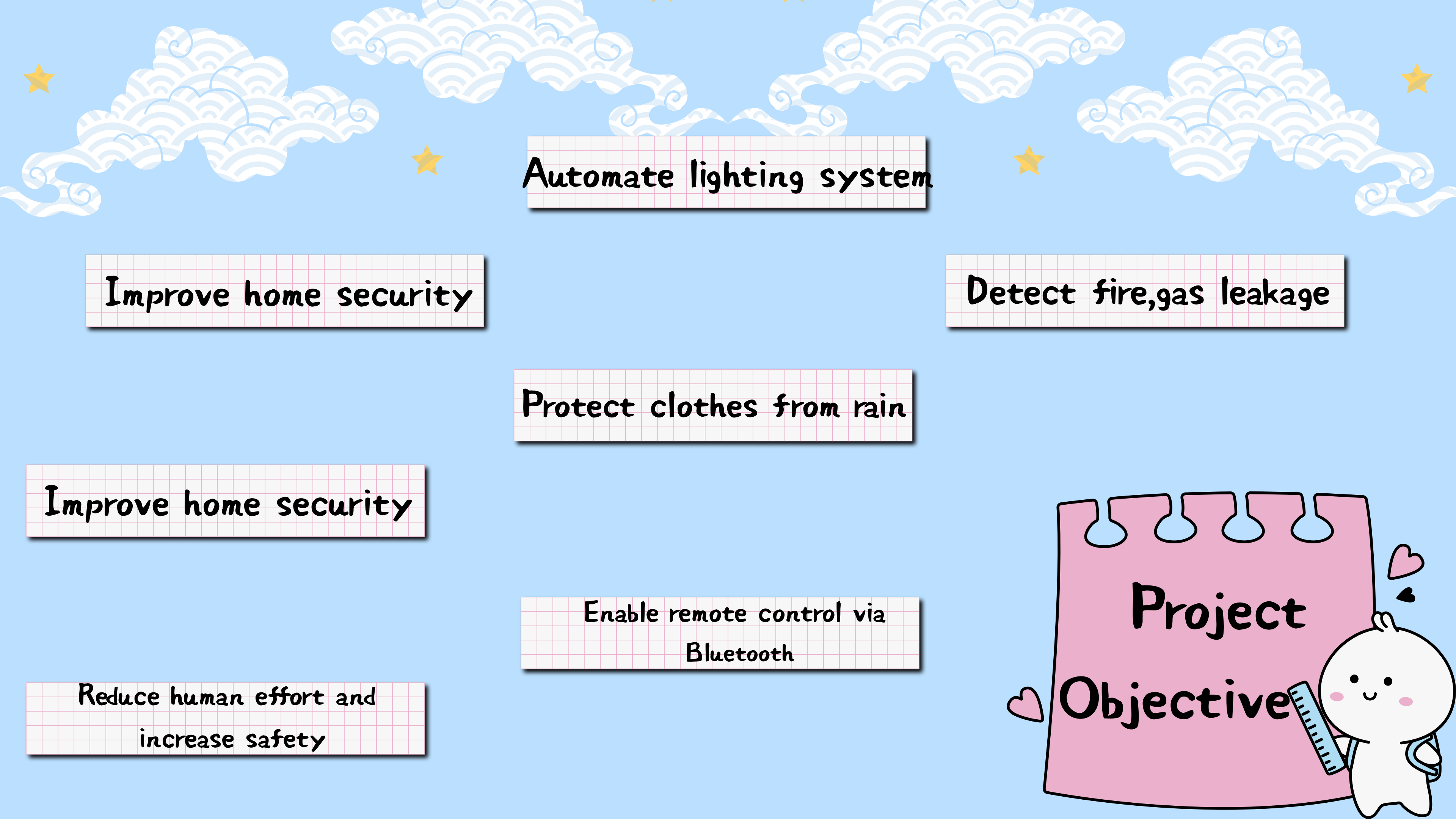






With the rapid development of technology, smart systems have become essential in modern life. Smart homes provide security, comfort, and energy efficiency.

Our project aims to design and implement a Smart Secure Home System using Arduino that enhances safety and allows remote control through a mobile phone.



Automate lighting system

Improve home security

Detect fire, gas leakage

Protect clothes from rain

Improve home security

Enable remote control via
Bluetooth

Reduce human effort and
increase safety



Problem Statement

Many homes suffer from:

- Weak security systems
- Gas leakage accidents
- Fire hazards
- Energy waste due to lights left on
- Clothes damage due to sudden rain
- No remote access to home systems



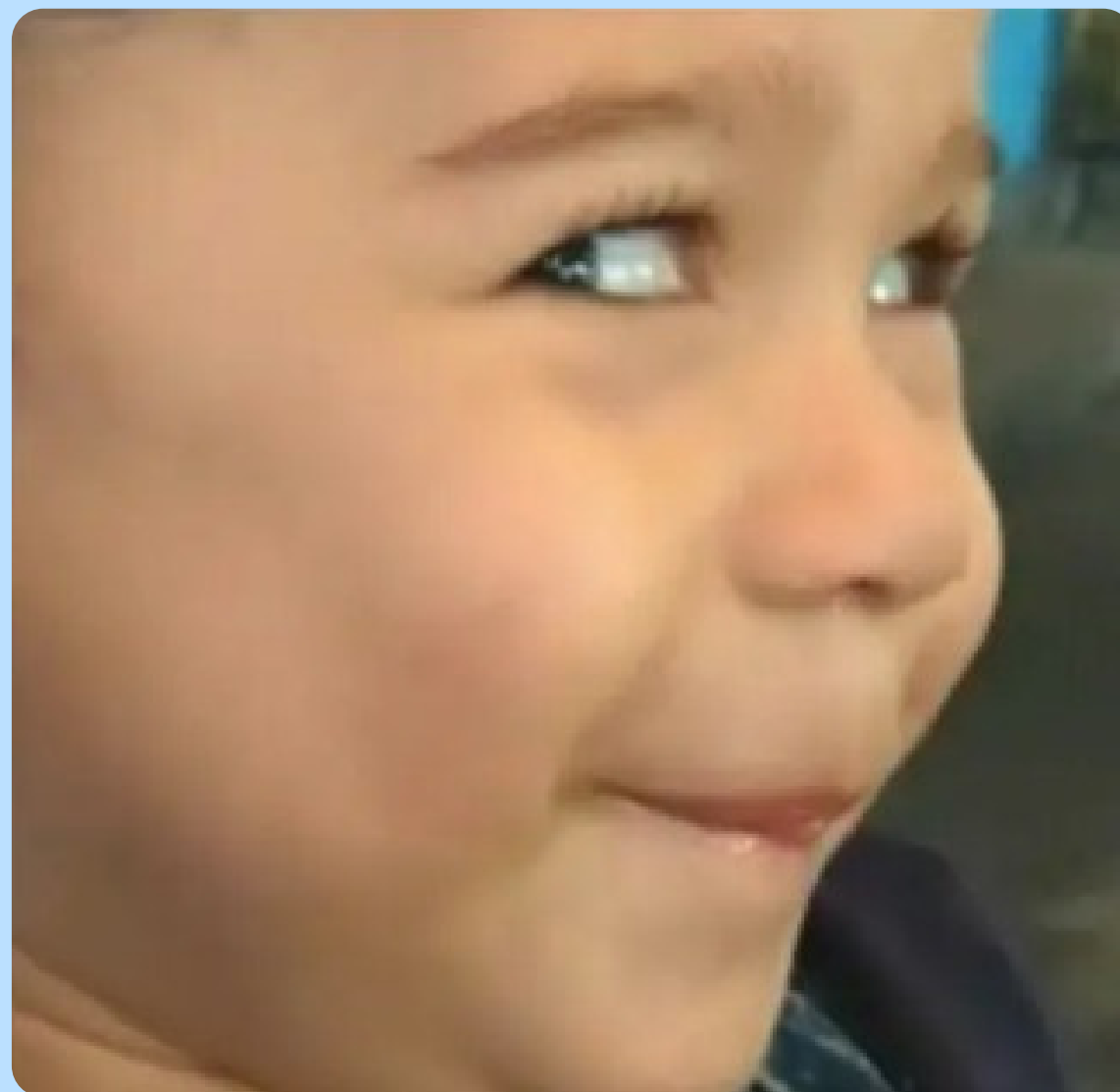
Proposed Solution



We propose a Smart Home System based on Arduino that includes:

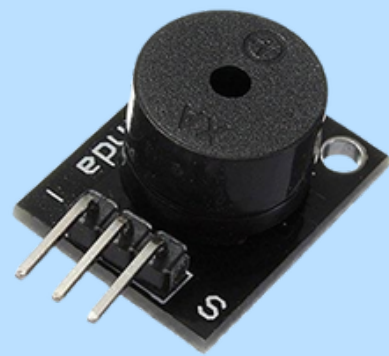
- Password-based door access
- Automatic rain detection system
- Motion-controlled lighting
- Fire detection system
- Gas leakage detection system
- Bluetooth-based remote control

مستعدین ندخل البيت ونشوف بیشتغل ازاي؟

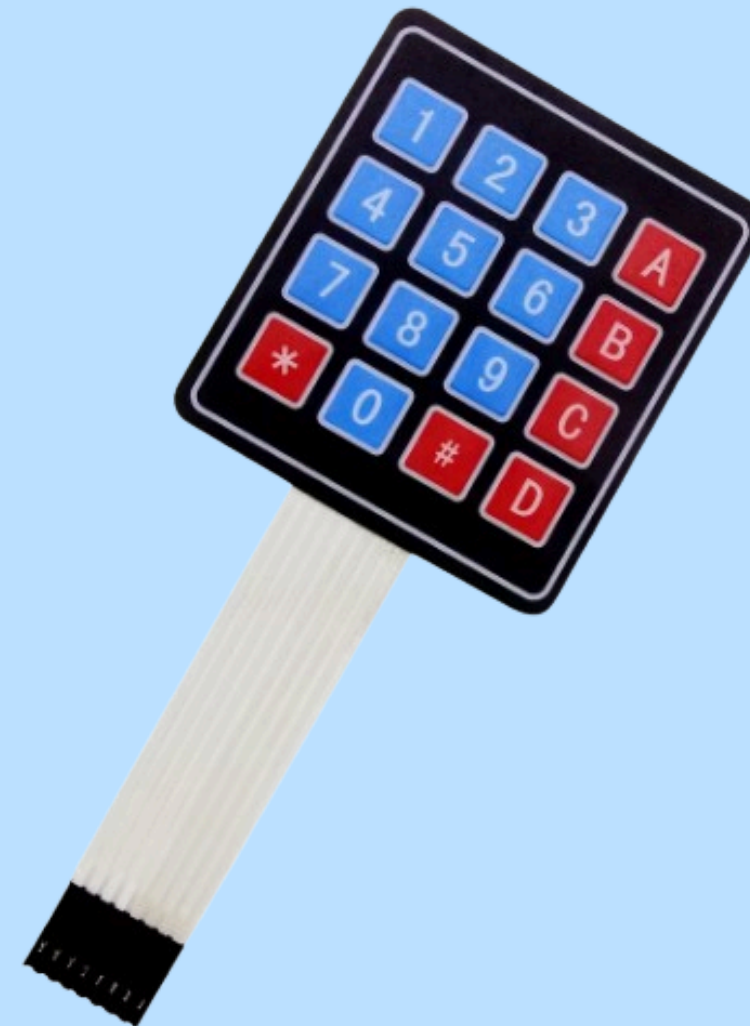


Gate

Buzzer Module



KeyPad



Gate

تفتكر هيحصل اي لو الباسورد غلط



Gate

Haramyyyyyyyyyyyyyyyyyy



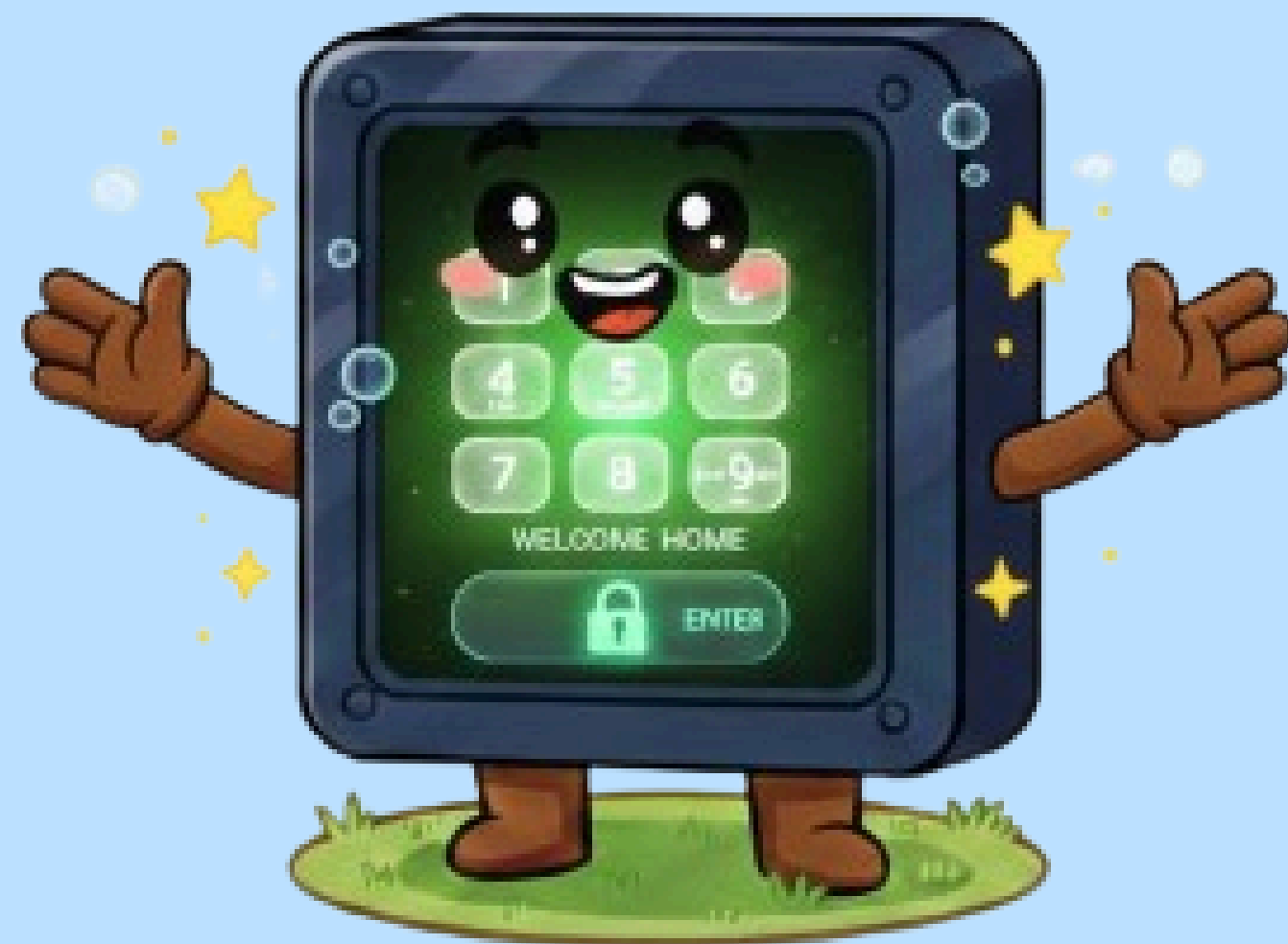
Gate

طيب لو صح؟



Gate

Gate Opened





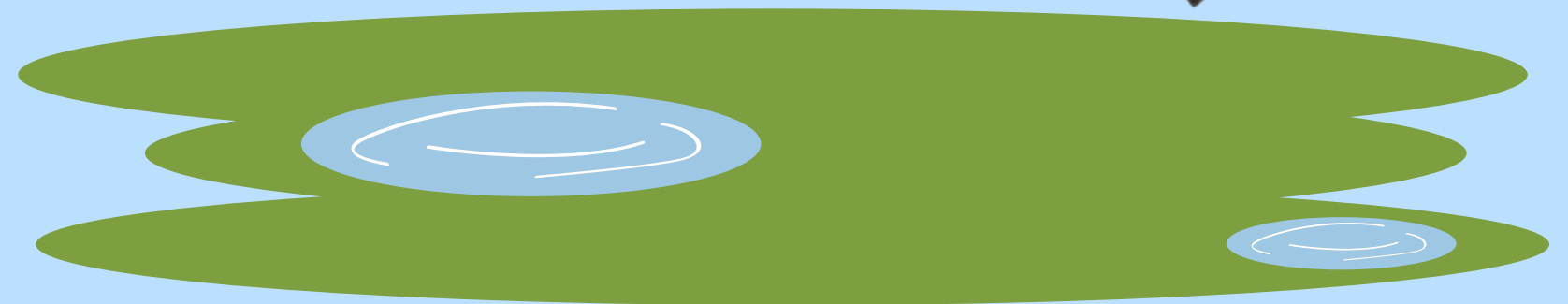
هي الدنيا مطرت
على بدلتي انا
بس و لا ايه؟!!

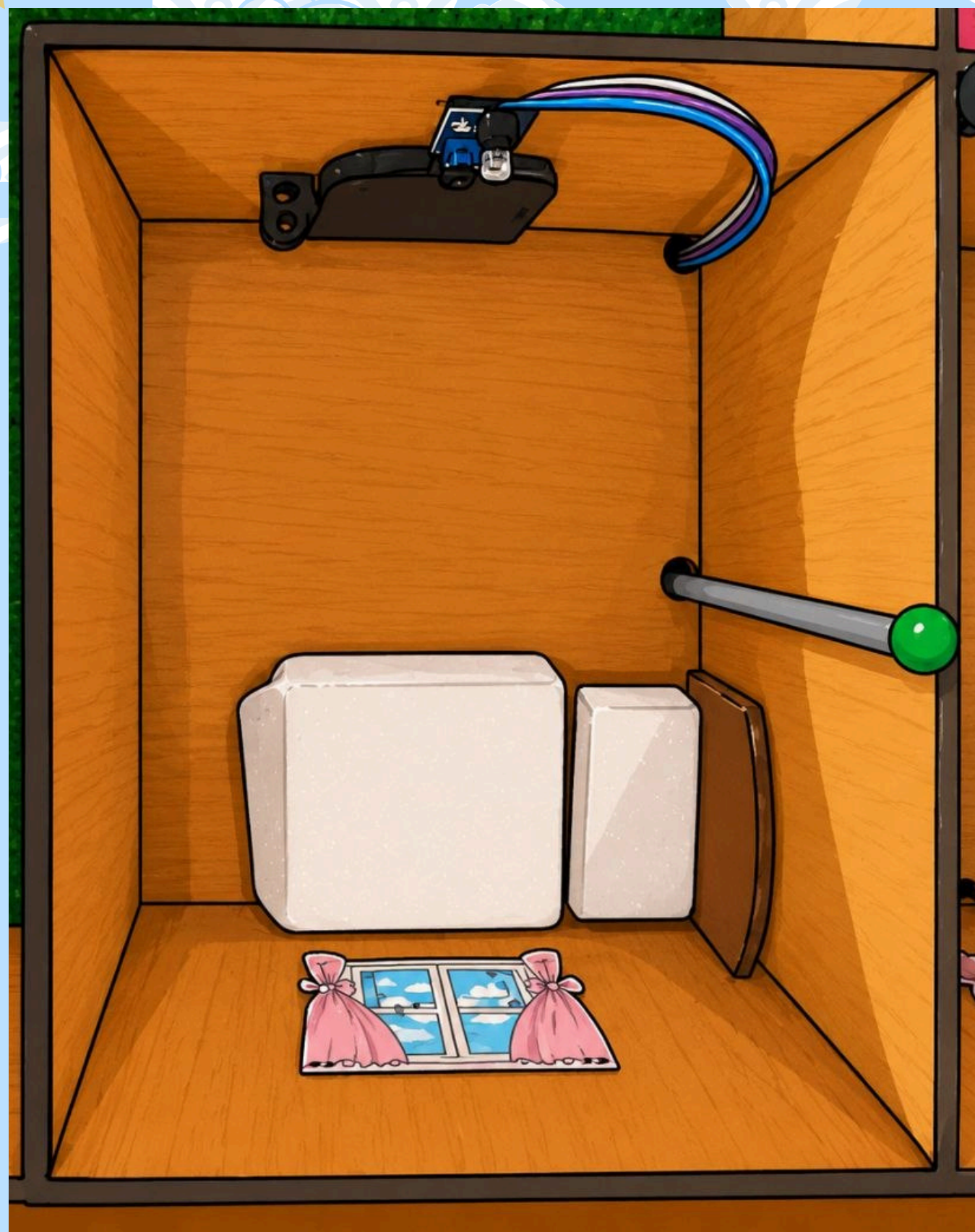


Garden

Servo

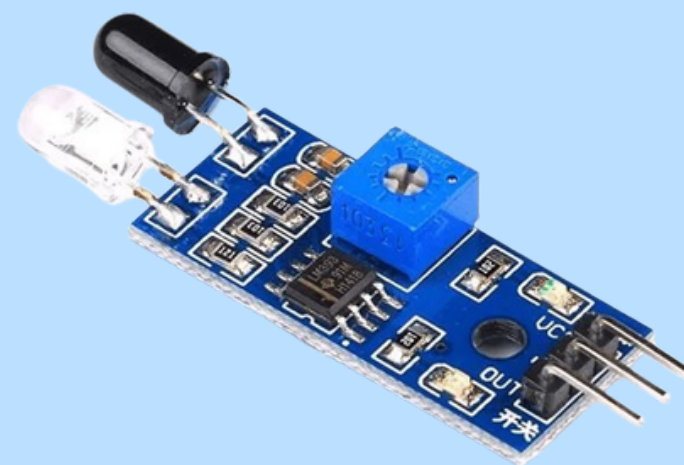
Rain Sensor





Room 1[★]

IR



LED

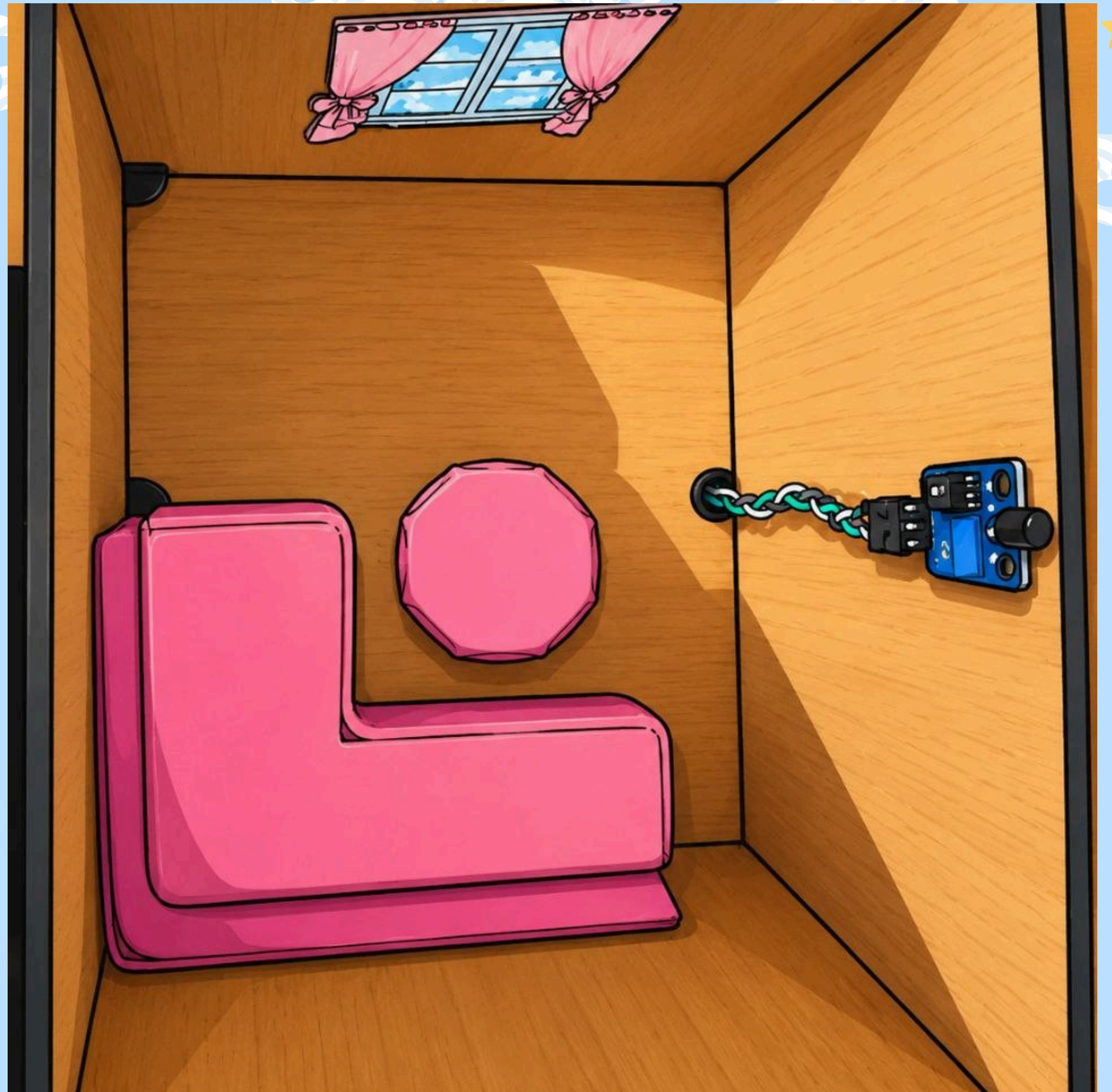
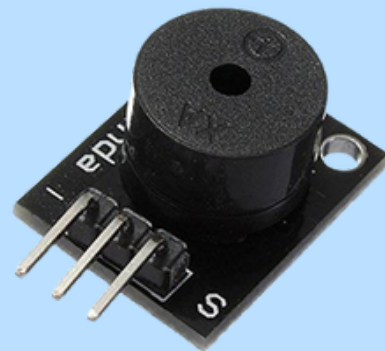


Room 2

Flame Sensor



Buzzer Module

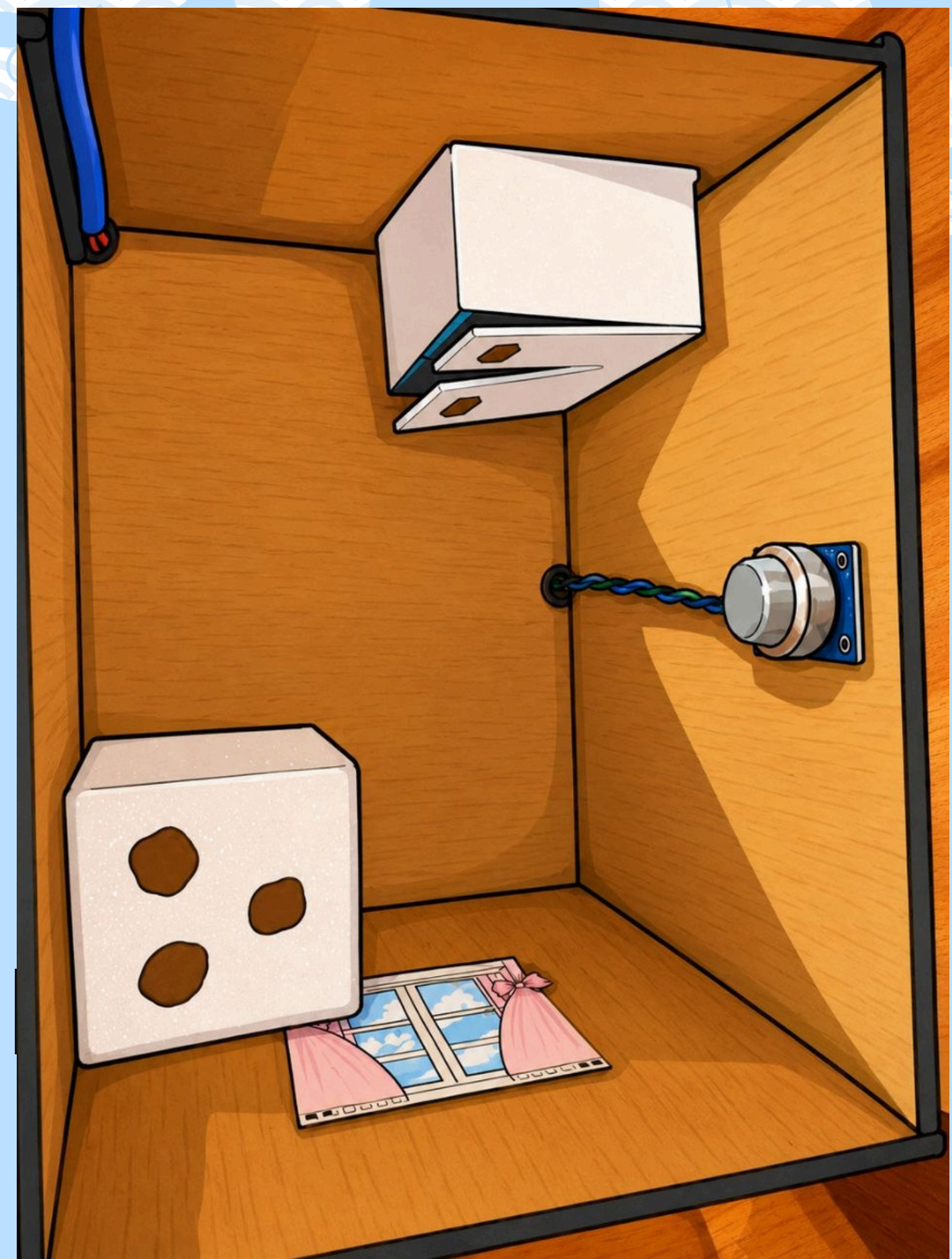
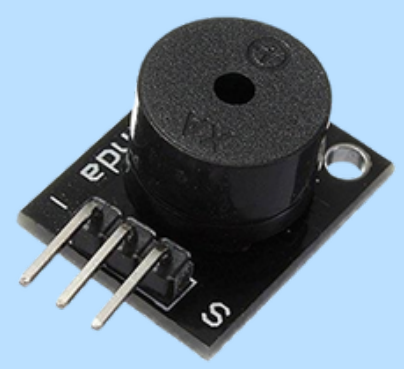


★ Room 3 ★

Gas Sensor

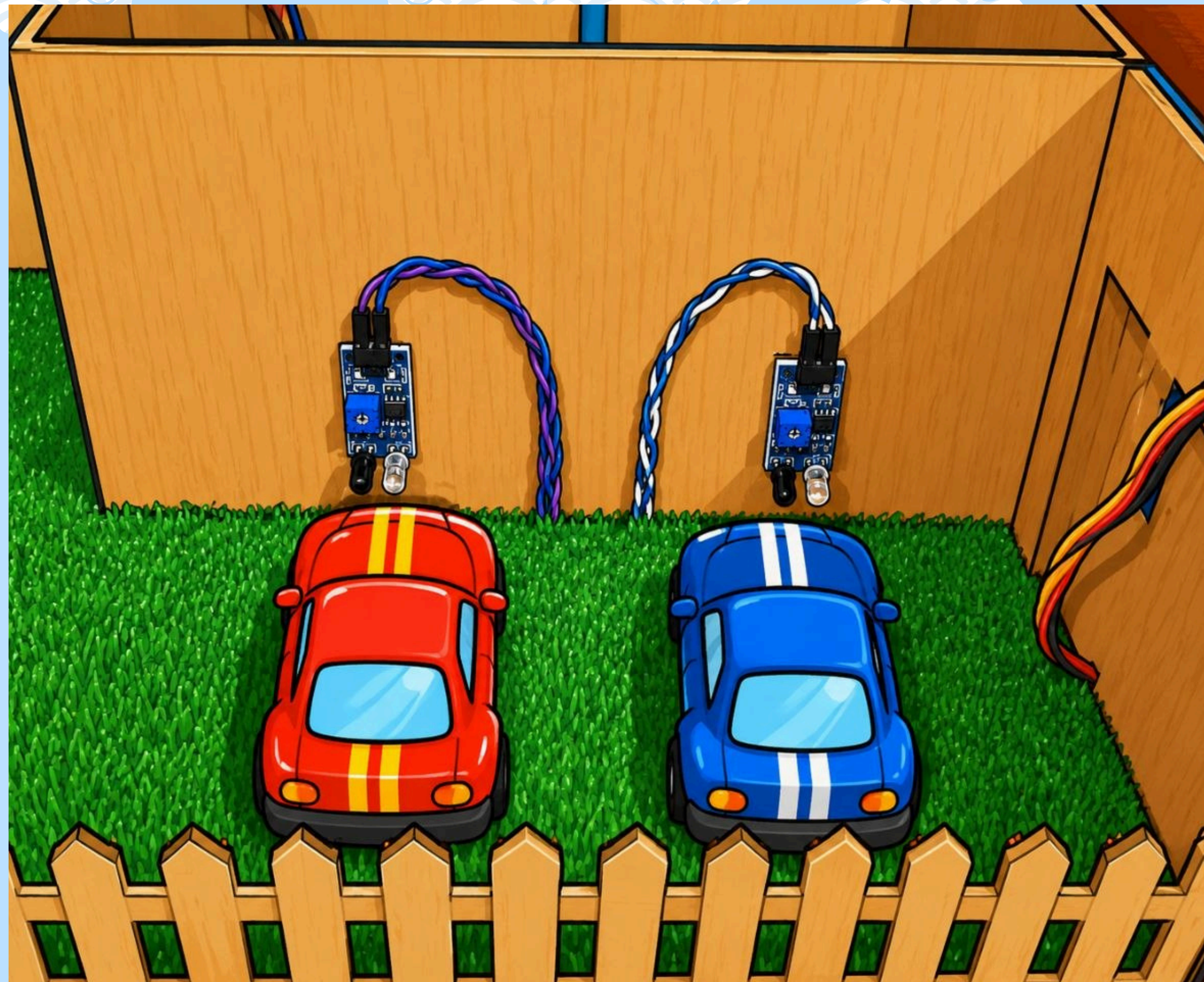
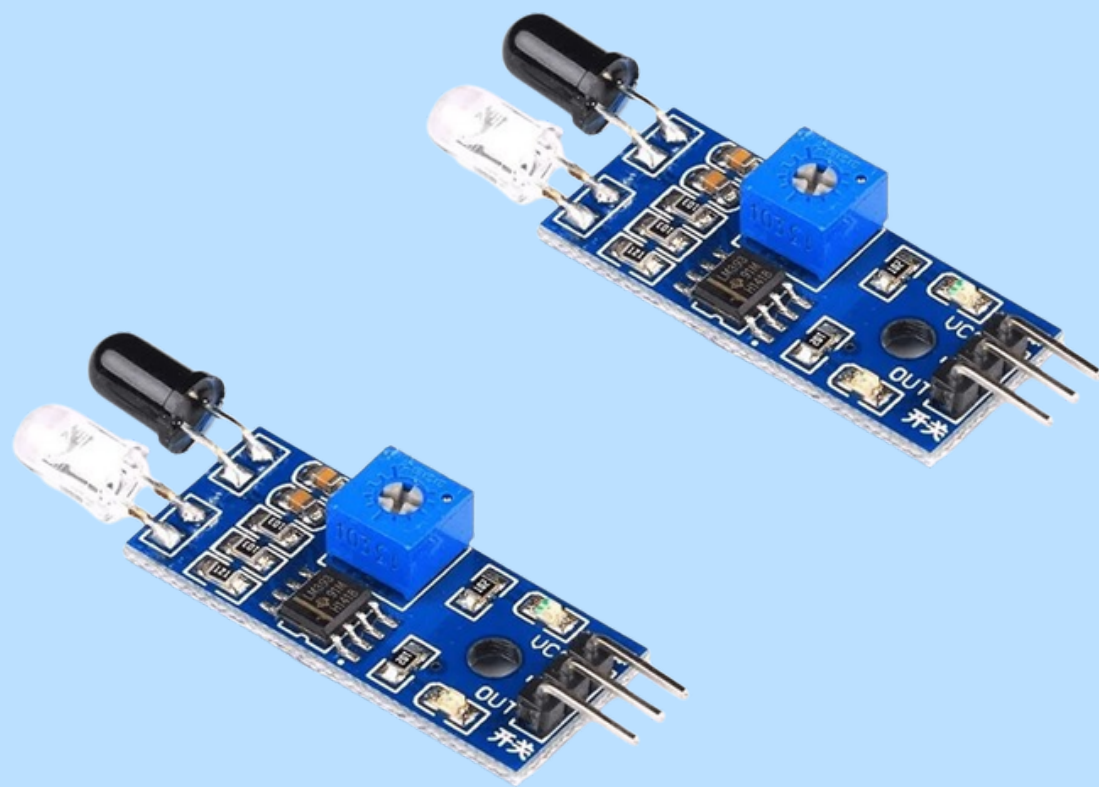


Buzzer Module



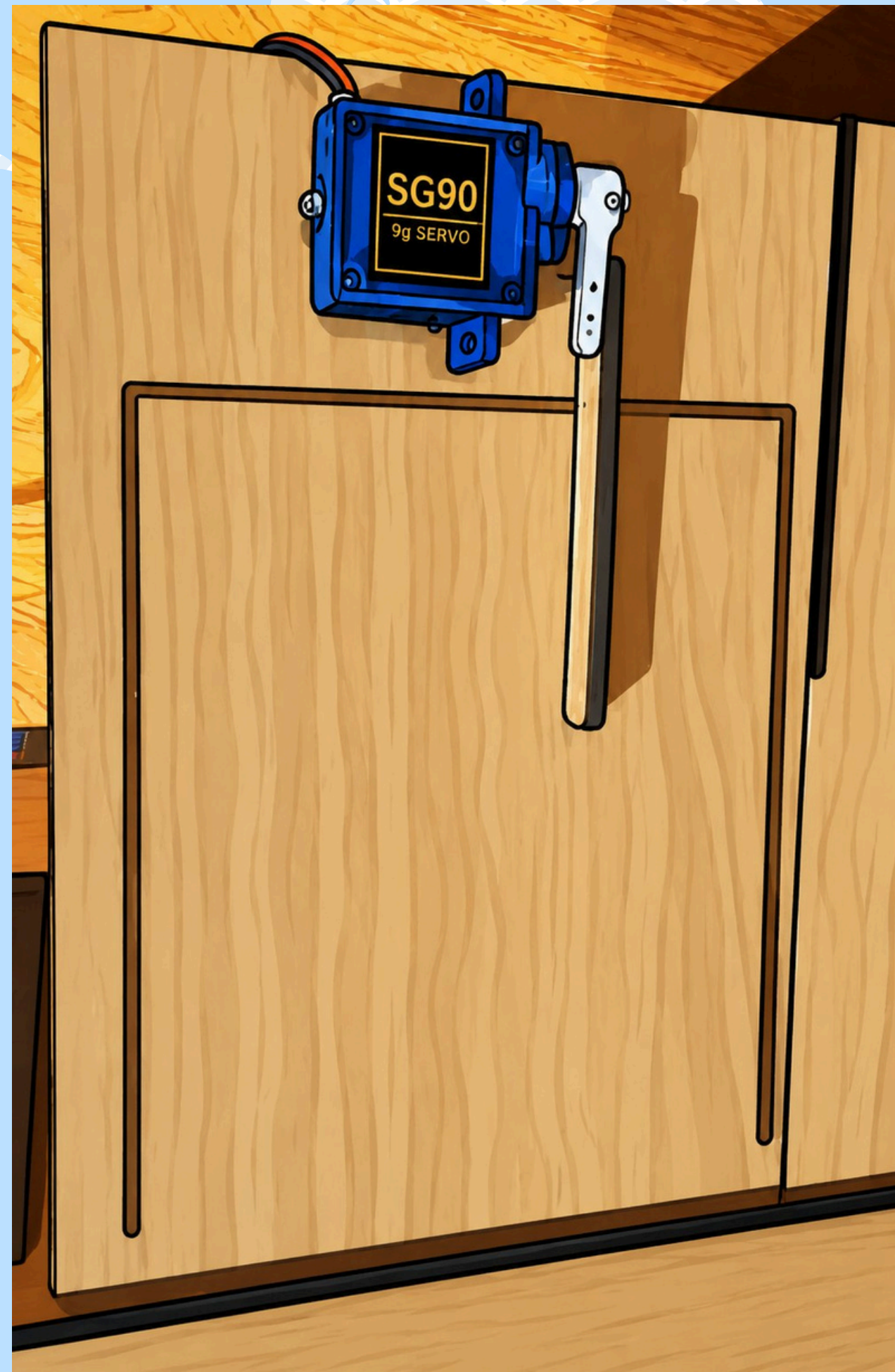
Garage

2 IR Sensors



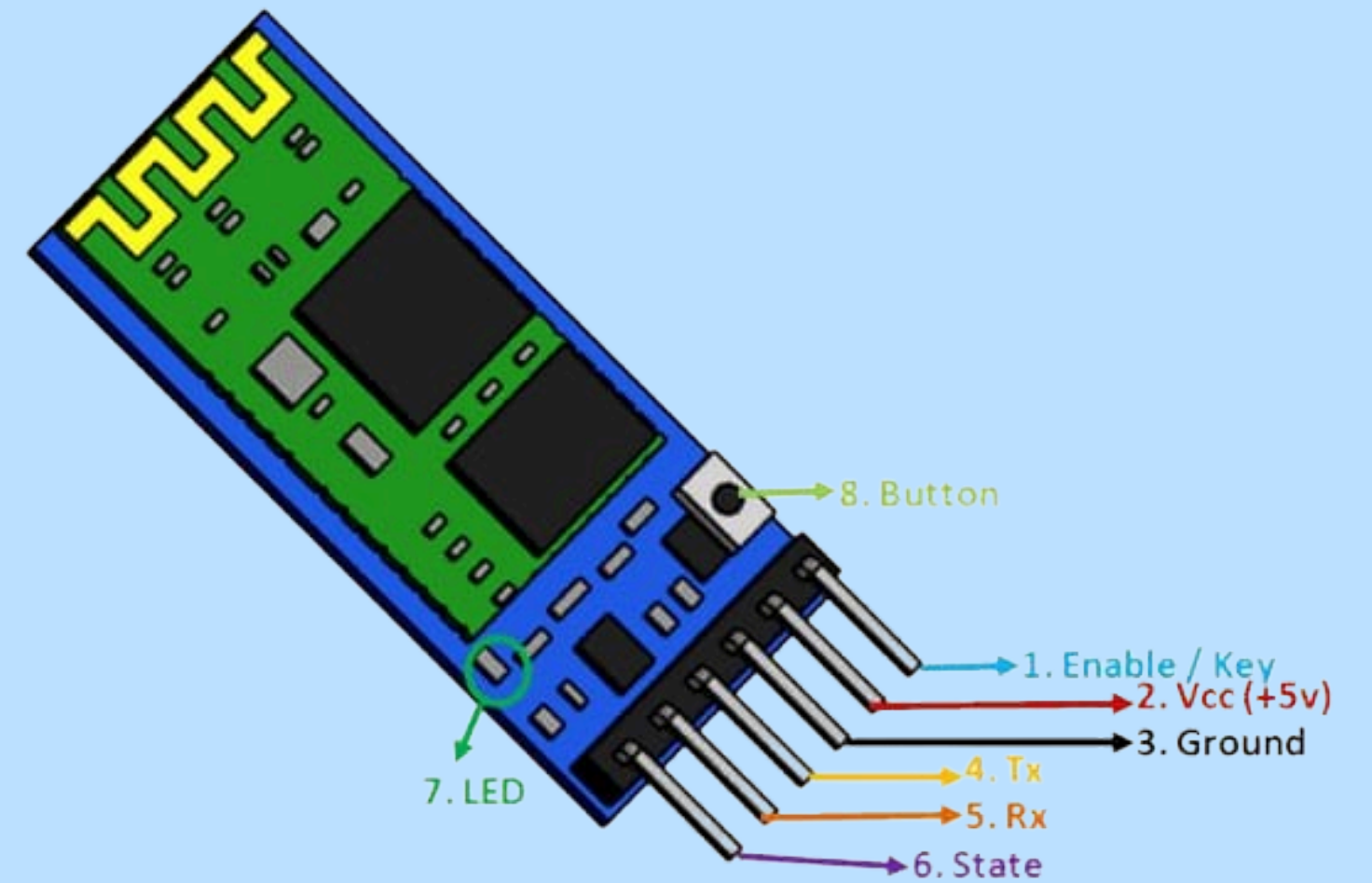
Gate

Servo





★ Voice Commands ★





★ Number of Functions ★

Password Authentication

LCD Feedback

Alarm after 3 wrong attempts

Door Control (Servo)

Rain Detection

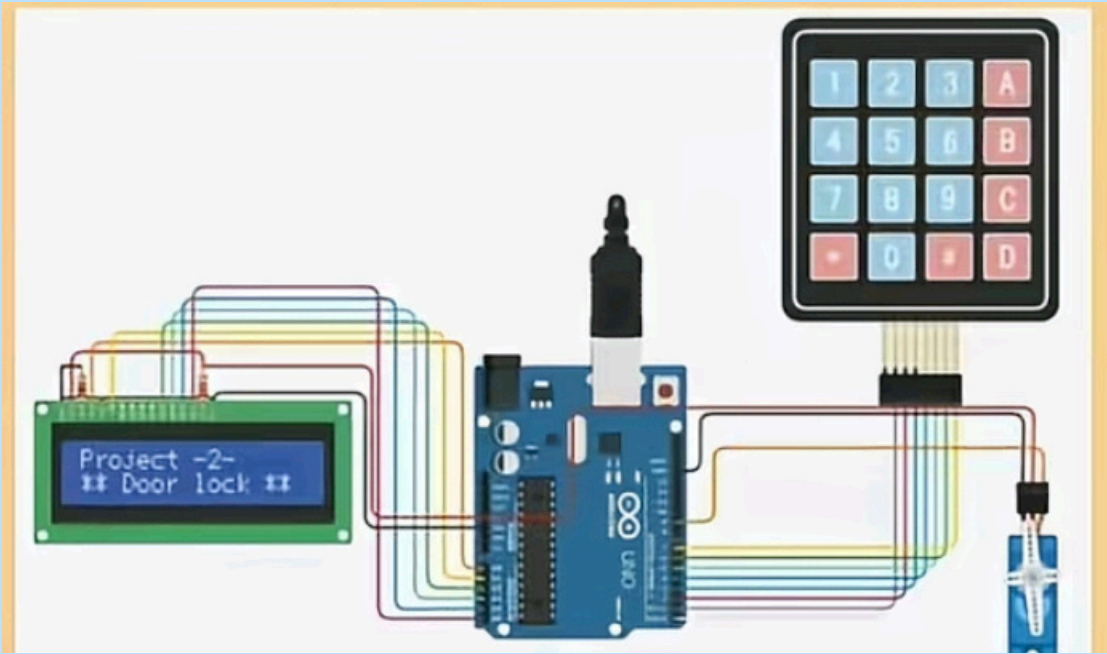
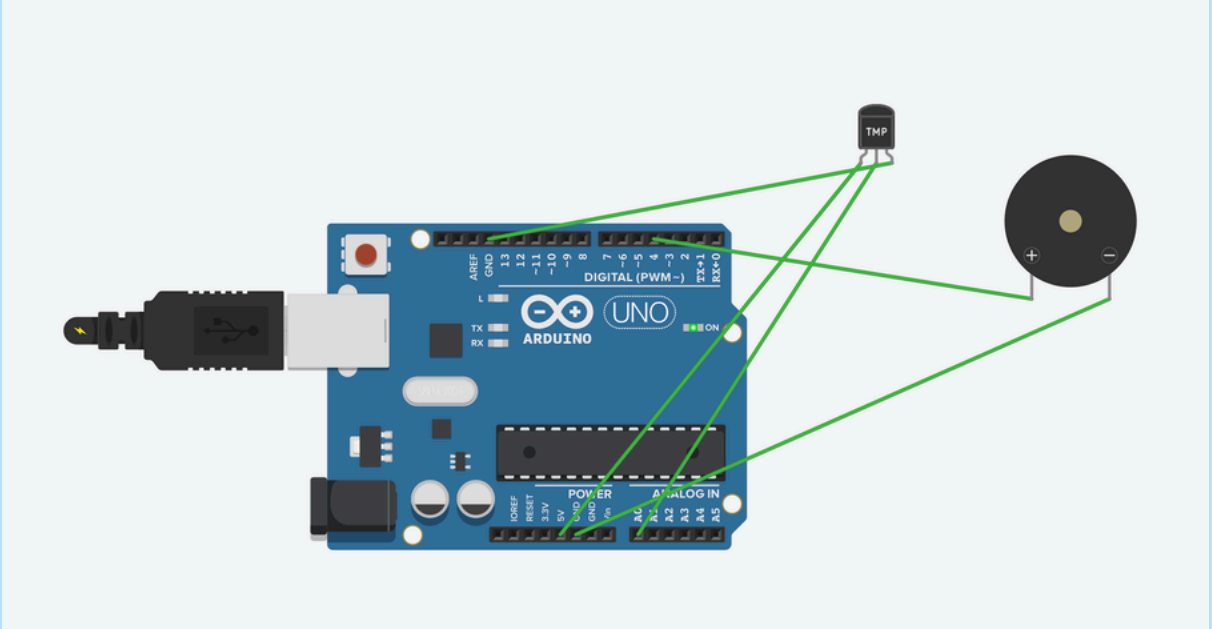
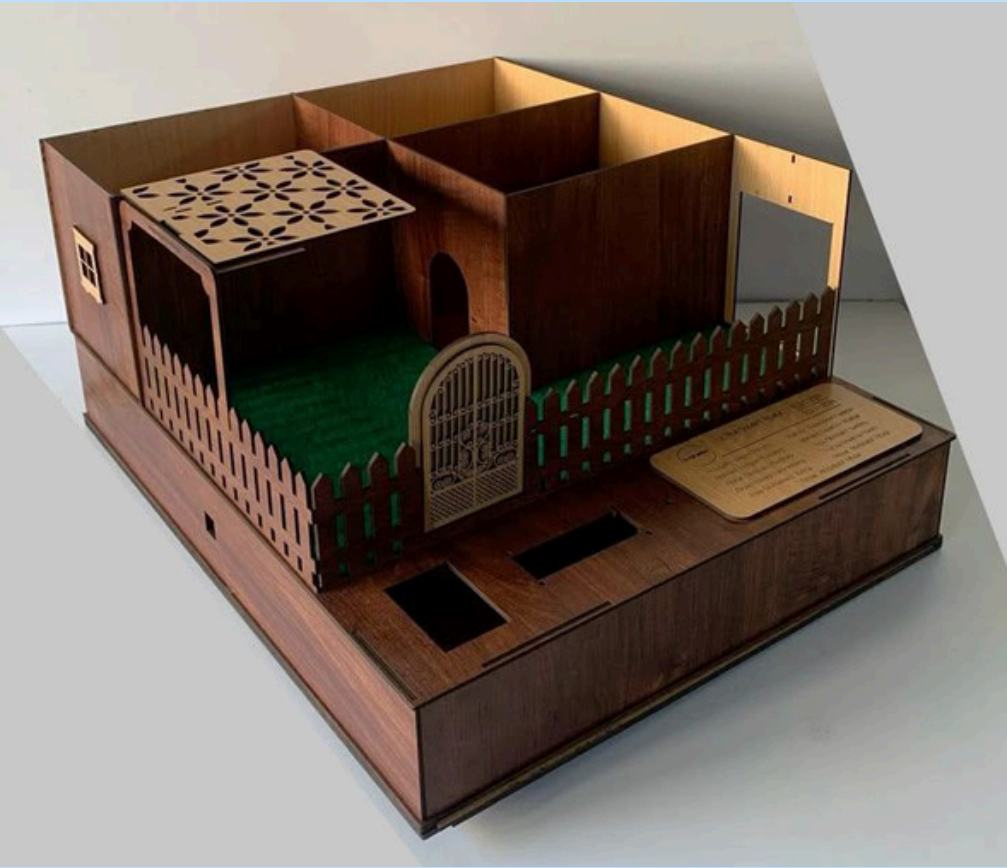
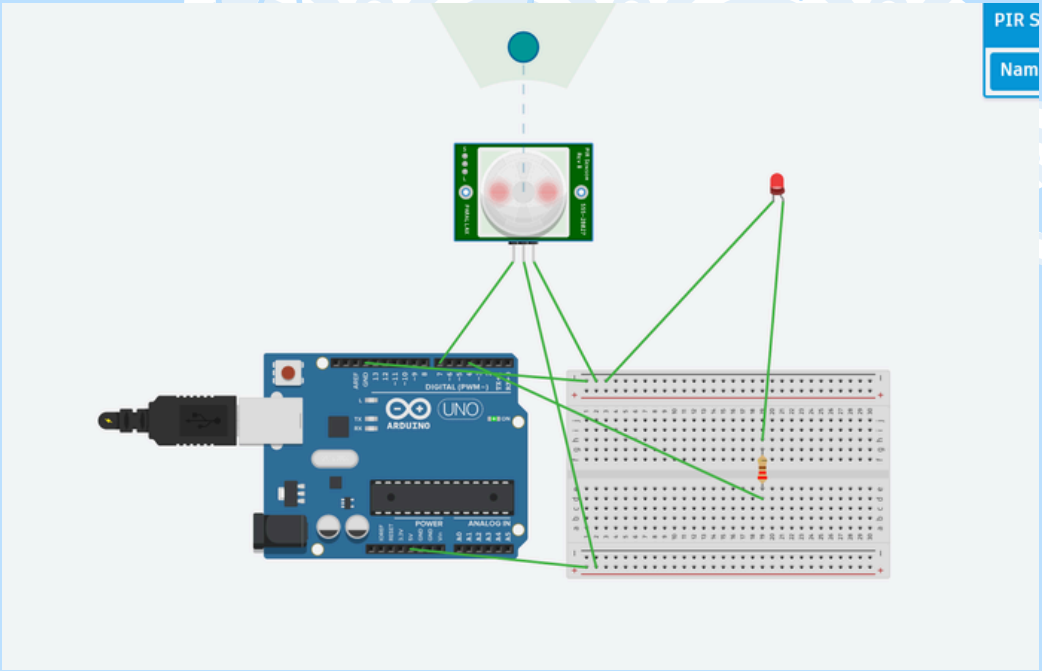
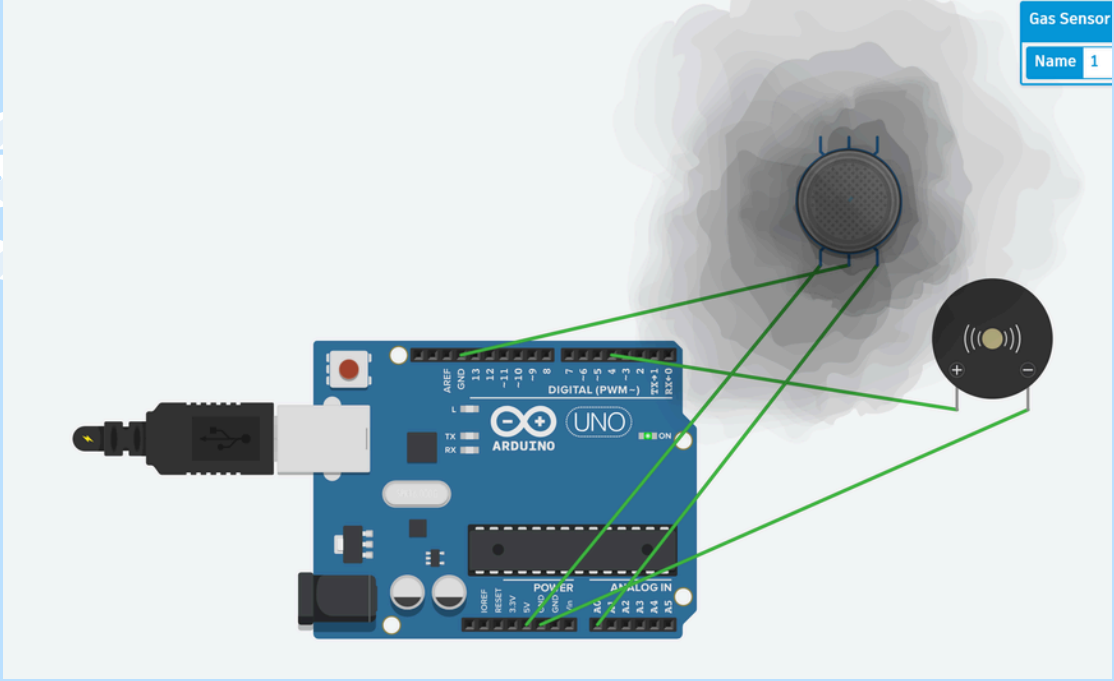
Clothesline Auto Movement

Motion-Based Lighting

Fire Detection

Gas Detection

Bluetooth Control



"The Journey Has Just Started!"

This project represents Version 1.0 — a successful proof of concept focused on safety and energy efficiency.

Our next vision is to expand from a Smart Home to a connected Smart City by integrating:

- AI-driven energy management
- IoT-based traffic and waste systems
- Centralized emergency response
- Smart public lighting and monitoring

Version 1.0 is the foundation for a smarter, safer future.

